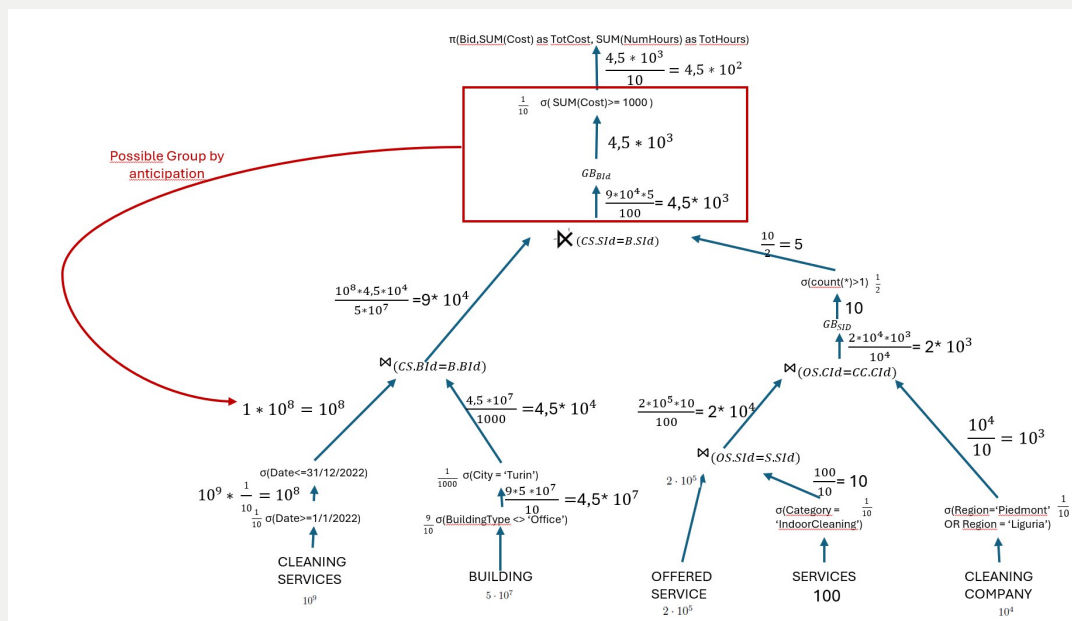


homework_3

Task 1

Report the corresponding algebraic expression and specify the cardinality of each node (representing an intermediate result or a leaf). If necessary, assume a data distribution. Also analyze the GROUP BY anticipation.

Since we don't have information about the distribution, we are assuming a uniform distribution.



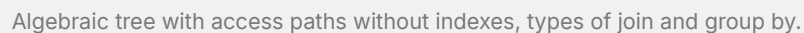
The anticipation of the **Group By BID with HAVING SUM(Cost) ≥ 1000** is optimal after the two filter operations on **Cleaning Services** due to the following factors:

- **Building** contains distinct rows with respect to **BID**
- Anticipating immediately after the join operation on **BID** would decrease efficiency

Regarding the **Group By SID with HAVING COUNT(*) > 1**, anticipation is not recommended because:

- **Services** contains distinct rows with respect to **SID**
- **Cleaning Company** lacks the **SID** attribute
- Grouping either **Offered Service** or the joined table on **SID** would cause the loss of essential **CID** values needed for the later join operation

Select one or more secondary physical structures to increase query performance. Justify your choice and report the corresponding execution plan (join orders, access methods, etc.).



The join operations are implemented as follows:

- Both **GROUP BY** operations (on *BID* and *SID*) follow **Nested Loops**, therefore utilizing **Group By Hash**.



The following indexes have been implemented:

- **Cleaning Services** : Secondary B+Tree index on Date (RF: 1/10), enabling INDEX RANGE SCAN + ACCESS BY ROWID
- **Building** : Secondary Hash index on City (RF: 1/1000), enabling INDEX RANGE SCAN + ACCESS BY ROWID
- **Offered Service** : Primary B+Tree index on CID, SID , enabling FULL SCAN INDEX
- **Services** : Secondary Hash index on Category (RF: 1/10), enabling INDEX RANGE SCAN + ACCESS BY ROWID
- **Cleaning Company** : Secondary Hash index on Region (RF: 1/10), enabling INDEX RANGE SCAN + ACCESS BY ROWID